Xuning Hu

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Education

Xi'an Jiaotong-Liverpool University (XJTLU) 2022/09 – 2026/06 (Expected)
BSc in Information and Computing Science Suzhou, China

- GPA: 3.72/4.00

Publications

[CHI-LBW'25] **X. Hu**, Y. Zhang, Y. Wei, Yue. Li, W. Stuerzlinger, H. N. Liang. "Exploring and Modeling Gaze-Based Steering Behavior in Virtual Reality." *2025 ACM CHI Conference on Human Factors in Computing Systems.* DOI: 10.1145/3706599.3720273 **Accepted**

[IEEE VR'25] **X. Hu**, X. Wen, Y. Wei, H. Zhang, J. Huang, H. N. Liang. "Optimizing Moving Target Selection in VR by Integrating Proximity-Based Feedback Types and Modalities." *The 32nd IEEE Conference on Virtual Reality and 3D User Interfaces.* DOI: 10.1109/VR59515.2025.00030 **Accepted**

[IEEE VR'25] X. Wen, Y. Wei, **X. Hu**, W. Stuerzlinger, Y. Wang, H. N. Liang. "Predicting Ray Pointer Landing Poses in VR Using Multimodal LSTM-Based Neural Networks." *The 32nd IEEE Conference on Virtual Reality and 3D User Interfaces.* DOI: 10.1109/VR59515.2025.00034 **Accepted**

[ISMAR'24] **X. Hu**, X. Yan, Y. Wei, W. Xu, Y. Li, Y. Liu, H. N. Liang. "Exploring the Effects of Spatial Constraints and Curvature for 3D Piloting in Virtual Environments." *2024 IEEE International Symposium on Mixed and Augmented Reality.* DOI: 10.1109/ISMAR62088.2024.00065 **Accepted**

[SUI'24] R. Darbar, **X. Hu**, X. Yan, Y. Wei, H. N. Liang, W. Xu, & S. Sarcar. "OnBodyQWERTY: An Empirical Evaluation of On-Body Tap Typing for AR HMDs." *2024 ACM Symposium on Spatial User Interaction.* DOI: https://dl.acm.org/doi/10.1145/3677386.3682084 [10.1145/3677386.3682084] **Accepted**

[SI3D'24] B. Chen, X. Yan, **X. Hu**, D. Kao, H. N. Liang. "Impact of Tutorial Modes with Different Time Flow Rates in Virtual Reality Games." *2024 ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games.* DOI: https://dl.acm.org/doi/10.1145/3651296 [10.1145/3651296] **Accepted**

[PACMHCI'24] R. Shi, Y. Wei, **X. Hu**, Y. Liu, Y. Li, L. Yu,H. N. Liang. "Experimental Analysis of Freehand Multi-Object Selection Techniques in Virtual Reality Head-Mounted Displays." *2024 Proceedings of the ACM on Human-Computer Interaction.* DOI: https://dl.acm.org/doi/abs/10.1145/3698129 [10.1145/3698129] **Accepted**

[TVCG'25] **X. Hu**, Y. Zhang, Y. Wei, Y. Li, W. Stuerzlinger, H. N. Liang. "Exploring and Modeling the Effects of Eye-Tracking Accuracy and Precision on Gaze-Based Steering in Virtual Environments." *2025 IEEE International Symposium on Mixed and Augmented Reality.* Under Review

[TVCG'25] **X. Hu**, Y. Qi, X. Yan, X. Wen, H. N. Liang, J. Huang. "Exploring Freehand-Based Selection Techniques of Polyhedra Faces in VR Environments" *2025 IEEE International Symposium on Mixed and Augmented Reality.*Under Review

[UIST'25] H. Zhang, Y. Xiao, J. Huang, X. Yan, **X. Hu**, Y. Zheng, N. Li, H. Tu, F. Tian "N-ary Gaussian Model: Modeling Pointing Uncertainty Across Task Scenarios Using an Automated Multi-Gaussian Modeling Pipeline." *2025 ACM Symposium on User Interface Software and Technology.* Under Review

PATENTS

Interactive control methods, devices, equipment, and storage media for VR game tutorials. CN118416465A (Published)

Research Experience

Simon Fraser University | VVISE Lab 2024/12 – present
Advisor: Prof. Wolfgang Stuerzlinger
Research Intern

The Hong Kong University of Science and Technology | Computational Media Lab 2024/10 – present
Advisor: Prof. Hai-Ning Liang
Research Assistant

Chinese Academy of Sciences | The Institute of Software 2024/06 – 2024/10
Advisor: Prof. Jin Huang
Research Assistant

Xi'an Jiaotong-Liverpool University | X-CHI Lab 2023/05 – 2024/02

Advisor: Prof. Hai-Ning Liang
Research Assistant

Selected Projects

Modeling the Impact of Eye-Tracking Accuracy and Precision on Gaze-Based Steering Tasks 2024/10 – 2025/01
Advisor: Prof. Hai-Ning Liang

- Developed a VR steering task system in Unity with Meta Quest Pro, focusing on gaze-based input interaction.
- Reconstructed the classical Steering Law to account for instability and length-insensitivity issues typical of gaze input, improving prediction accuracy
- Collected users' **spatial precision and spatial accuracy errors** from eye tracking as baseline noise, simulating precision with fixed offsets and accuracy with white Gaussian noise
- Quantified how different spatial error types affect task performance, fitted a modified steering model to **improve interaction robustness under degraded tracking quality**
- The study has been accepted to **CHI-LBW '25** and submitted to **IEEE TVCG '25**

Optimizing Moving Target Selection in VR with Proximity-Based Feedback 2024/07 – 2024/10
Advisor: Prof. Hai-Ning Liang

- Designed feedback mechanisms based on the Proximity mechanism for **visual**, **auditory**, and **haptic** modalities, categorizing feedback types into **Binary**, **Continuous**, and **Partial**.
- In **Study 1**, we explored the impact of feedback types and feedback mechanisms on the selection of moving objects across different modalities, using metrics such as time, error rate, UEQ ranking, and interviews. In **Study 2**, we further investigated the effects of different modality combinations on the selection task.
- The study has been accepted to **IEEE VR'25**

Modeling User Behaviors For Steering Law in Virtual Reality 2024/01 – 2024/04
Advisor: Prof. Hai-Ning Liang

- Developed a **virtual drone control system** in Unity using Oculus and XBOX controllers. The system employed **Mesh computation** to create paths, incorporating specific constraints on width, height, and curvature.
- Tested the adaptability of both the **original Steering Law and the weighted Euclidean version under dual-axis width constraints in a 3D environment (Steering Law: $R^2 = 0.923$; weighted Euclidean version: $R^2 = 0.968$)**.
- To investigate the effect of the weight coefficient η in the weighted Euclidean model under complex scenarios, we introduced seven curvature-constrained paths. Our findings indicated that the change in the weight coefficient followed a normal distribution curve relative to the curvature radius **(η: $R^2 = 0.84$)**. By combining the weighted Euclidean model with the curvature model and replacing the constant term with a normal distribution function influenced by curvature, **our new model demonstrated a 52.6% improvement in AIC and a 60.6% improvement in R^2 compared to the baseline Euclidean model**.

Technical Skills

Programming and Tools: C#, C, C++, Java, Python, Matlab, Unity, Meta Interaction SDK, Hololens, Vicon, SPSS
Tech and Soft Skills: Motion Capture, XR development, Modeling User Behavior, User Study Design, Eye-Tracking Data Analysis Algorithm, Reinforcement learning

Extra-Curriculum Experience

Oral Presentation 32nd IEEE Conference on Virtual Reality and 3D User Interfaces (VR) 2025/03
Oral Presentation 24nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2024/10
Student Volunteer 23nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2023/10
China Undergraduate Mathematical Contest in Modelling 2023/09
RoboMaster Robotics Competition : National Second prize in RoboMaster 2023 2023/03